Quiz 12 Solution.

Q. Write a C++ program which takes an integer as input and display whether it is a prime number or not. (Using recursion).

Sol.

```
# include <iostream>
using namespace std;
```
⇒ 1 mark

```
bool isRecursion (int n, int i = 2) {
        if (n <= 1) {
                return false;
        }
```
⇒ 1 mark (for handling of negative numbers)

```
    else if (i >= n/2) {
            return true;
    }
```
⇒ 2 marks

```
    else if (n % i == 0) {
            return false;
    }
```
⇒ 2 marks

```
    return isRecursion (n, i+1);
```
⇒ 2 marks

```
}

int main() {
        int n;
        cout << " Enter number : ";
        cin >> n;
```
⇒ 1 mark

```
    if ( isRecursion (n) ) cout << "Entered no. is prime";
    else cout << " Not a prime no.";
```
⇒ 1 mark

```
return 0; }
```